D. GCD-sequence

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: greedy, implementation, math, number theory | https://codeforces.com/problemset/problem/1980/D

GCD (Greatest Common Divisor) of two integers x and y is the maximum integer z by which both x and y are divisible. For example, *GCD*(36, 48) = 12, *GCD*(5, 10) = 5, and *GCD*(7,11) = 1.

Kristina has an array a consisting of exactly n positive integers. She wants to count the GCD of each neighbouring pair of numbers to get a new array b, called *GCD-sequence*.

So, the elements of the GCD-sequence b will be calculated using the formula $b_i = GCD(a_i, a_{i+1})$ for $1 \le i \le n - 1$.

Determine whether it is possible to remove **exactly one** number from the array a so that the GCD sequence b is non-decreasing (i.e., $b_i \le b_{i+1}$ is always true).

For example, let Khristina had an array a = [20, 6, 12, 3, 48, 36]. If she removes $a_4 = 3$ from it and counts the GCD-sequence of b, she gets:

- $b_1 = GCD(20, 6) = 2$
- $b_2 = GCD(6, 12) = 6$
- $b_3 = GCD(12, 48) = 12$
- $b_4 = GCD(48, 36) = 12$

The resulting GCD sequence b = [2,6,12,12] is non-decreasing because $b_1 \le b_2 \le b_3 \le b_4$.

Input

The first line of input data contains a single number t ($1 \le t \le 10^4$) — he number of test cases in the test.

This is followed by the descriptions of the test cases.

The first line of each test case contains a single integer n ($3 \le n \le 2 \cdot 10^5$) — the number of elements in the array a.

The second line of each test case contains exactly n integers a_i ($1 \le a_i \le 10^9$) — the elements of array a.

It is guaranteed that the sum of n over all test case does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single line:

- "YES" if you can remove **exactly one** number from the array a so that the GCD-sequence of b is non-decreasing;
- "NO" otherwise.

You can output the answer in any case (for example, the strings "yEs", "yes", "Yes", and "YES" will all be recognized as a positive answer).

Example

Input

```
12
6
20 6 12 3 48 36
4
12 6 3 4
3
10 12 3
5
32 16 8 4 2
5
100 50 2 10 20
4
2 4 8 1
10
7 4 6 2 4 5 1 4 2 8
7
5 9 6 8 5 9 2
6
11 14 8 12 9 3
9
5 7 3 10 6 3 12 6 3
3
4 2 4
8
1 6 11 12 6 12 3 6
```

Output

```
YES
NO
YES
NO
YES
YES
NO
YES
YES
YES
YES
YES
```

Note

The first test case is explained in the problem statement.